

Linux Assignment 4

Question 1: User Permissions and Shared Directories

A development team has three users: developer1, developer2 and tester1. Create a shared directory /project/application where developers can create, modify and delete files, but tester1 should have read-only access. Ensure that newly created files inherit the project's group ownership and users cannot delete other users' files.

```
[root@localhost folder1]# sudo usermod -aG developers dev1
[root@localhost folder1]# sudo usermod -aG developers dev2
[root@localhost folder1]# sudo mkdir -p /project/application
[root@localhost folder1]# sudo chown root:developers /project/application
[root@localhost folder1]# sudo chmod 3770 /project/application
[root@localhost folder1]# sudo setfacl -m u:tester1:rx /project/application
[root@localhost folder1]# ls -ld /project/application
drwxrws--T+ 2 root developers 6 Oct  2 13:52 /project/application
[root@localhost folder1]# getfacl /project/application
getfacl: Removing leading '/' from absolute path names
# file: project/application
# owner: root
# group: developers
# flags: -st
user::rwx
user:tester1:r-x
group::rwx
mask::rwx
other::---
```

Question 2: Log Management and Disk Space

A RHEL server is running out of disk space because /var/log/messages has grown significantly. Configure log rotation so that the log file is rotated daily, retains seven days of history, compresses older logs and creates a new log file with appropriate ownership and

```
[root@localhost folder1]# rpm -q logrotate
logrotate-3.22.0-4.el10.x86_64
[root@localhost folder1]# sudo vi /etc/logrotate.d/messages
[root@localhost folder1]# sudo logrotate -d /etc/logrotate.d/messages
warning: logrotate in debug mode does nothing except printing debug messages! Consider using verbose mode (-v) instead if this is not what you want.

reading config file /etc/logrotate.d/messages
Reading state from file: /var/lib/logrotate/logrotate.status
Allocating hash table for state file, size 64 entries
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Handling 1 logs

rotating pattern: /var/log/messages after 1 days empty log files are not rotated, (7 rotations), old logs are removed
considering log /var/log/messages
Now: 2026-10-02 15:26
Last rotated at 2026-09-29 22:11
log needs rotating
rotating log /var/log/messages, log->rotateCount is 7
dateext suffix '-20261002'
glob pattern '-[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9]'
set default create context to system_u:object_r:var_log_t:s0
renaming /var/log/messages.7.gz to /var/log/messages.8.gz (rotatecount 7, logstart 1, i 7),
renaming /var/log/messages.6.gz to /var/log/messages.7.gz (rotatecount 7, logstart 1, i 6),
renaming /var/log/messages.5.gz to /var/log/messages.6.gz (rotatecount 7, logstart 1, i 5),
renaming /var/log/messages.4.gz to /var/log/messages.5.gz (rotatecount 7, logstart 1, i 4),
renaming /var/log/messages.3.gz to /var/log/messages.4.gz (rotatecount 7, logstart 1, i 3),
renaming /var/log/messages.2.gz to /var/log/messages.3.gz (rotatecount 7, logstart 1, i 2),
renaming /var/log/messages.1.gz to /var/log/messages.2.gz (rotatecount 7, logstart 1, i 1),
log /var/log/messages.8.gz doesn't exist -- won't try to dispose of it
set default create context to system_u:object_r:var_log_t:s0
renaming /var/log/messages to /var/log/messages.1
creating new /var/log/messages mode = 0640 uid = 0 gid = 0
compressing log with: /bin/gzip
[root@localhost folder1]# sudo logrotate -f /etc/logrotate.d/messages
[root@localhost folder1]# ls -lh /var/log/messages*
-rw-r-----. 1 root root    0 Oct  2 15:28 /var/log/messages
-rw-----. 1 root root 39K Oct  2 15:28 /var/log/messages.1.gz
-rw-----. 1 root root 1.8M Sep 29 21:39 /var/log/messages-20260929
[root@localhost folder1]# ls -l /var/log/messages
-rw-r-----. 1 root root 0 Oct  2 15:28 /var/log/messages
[root@localhost folder1]#
```

A new 350MB disk, /dev/sda, has been attached to a Linux server. Create a 315 MB partition using LVM, configure it with XFS, mount it permanently at /data/application and ensure that it is automatically mounted after reboot. Leave the remaining disk space available for future use.

```
[root@localhost ~]# fdisk /dev/sda
```

Welcome to fdisk (util-linux 2.40.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): n

Partition number (6-128, default 6): 6

First sector (25792512-31457246, default 25792512):

Last sector, +/-sectors or +/-size{K,M,G,T,P} (25792512-31457246, default 31455231): +350M

Created a new partition 6 of type 'Linux filesystem' and of size 350 MiB.

Command (m for help): w

The partition table has been altered.
Syncing disks.

```
[root@localhost ~]# partprobe
```

Warning: Unable to open /dev/sr0 read-write (Read-only file system). /dev/sr0 has been opened read-only.

```
[root@localhost ~]# sync
```

```
[root@localhost ~]# lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	15G	0	disk	
├─sda1	8:1	0	1M	0	part	
├─sda2	8:2	0	976M	0	part	/boot
├─sda3	8:3	0	10G	0	part	/
├─sda4	8:4	0	976M	0	part	[SWAP]
├─sda5	8:5	0	400M	0	part	
└─┬appvg-applv	253:0	0	352M	0	lvm	/srv/appdata
└─sda6	8:6	0	350M	0	part	
sr0	11:0	1	51M	0	rom	

```
[root@localhost ~]# pvcreate /dev/sda6
```

Physical volume "/dev/sda6" successfully created.

```
[root@localhost ~]# pvs
```

PV	VG	Fmt	Attr	PSize	PFree
/dev/sda5	appvg	lvm2	a--	396.00m	44.00m
/dev/sda6		lvm2	---	350.00m	350.00m

```

[root@localhost ~]# vgcreate vgapp /dev/sda6
Volume group "vgapp" successfully created
[root@localhost ~]# vgs
VG      #PV #LV #SN Attr   VSize   VFree
appvg   1   1   0 wz--n- 396.00m 44.00m
vgapp   1   0   0 wz--n- 348.00m 348.00m
[root@localhost ~]# lvcreate -L 315M -n lvapp vgapp
Rounding up size to full physical extent 316.00 MiB
Logical volume "lvapp" created.
[root@localhost ~]# lvs
LV      VG      Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync
Convert
applv   appvg   -wi-ao---- 352.00m

lvapp   vgapp   -wi-a----- 316.00m

[root@localhost ~]# mkfs.xfs /dev/vgapp/lvapp
meta-data=/dev/vgapp/lvapp      isize=512    agcount=4, agsize=20224 blks
      =                       sectsz=512    attr=2, projid32bit=1
      =                       crc=1          finobt=1, sparse=1, rmapbt=1
      =                       reflink=1       bigtime=1 inobtcount=1 nrext64=
1
      =                       exchange=0
data    =                       bsize=4096   blocks=80896, imaxpct=25
      =                       sunit=0        swidth=0 blks
naming  =version 2             bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log        bsize=4096   blocks=16384, version=2
      =                       sectsz=512    sunit=0 blks, lazy-count=1
realtime =none                extsz=4096   blocks=0, rtextents=0
[root@localhost ~]# mkdir -p /data2/app2
[root@localhost ~]# mount /dev/vgapp/lvapp /data2/app2
[root@localhost ~]# df -h /data2/app2
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/vgapp-lvapp  252M      21M   232M   9% /data2/app2
[root@localhost ~]# blkid /dev/vgapp/lvapp
/dev/vgapp/lvapp: UUID="0fbffdf6-d7c3-47db-b0d0-fbfd82087898" BLOCK_SIZE="512"
" TYPE="xfs"
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Tue Aug 11 12:09:27 2026
#

```

```

# /etc/fstab
# Created by anaconda on Tue Aug 11 12:09:27 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=f902a1e3-4a9c-42dd-b32f-af27dd971fa4 /                xfs      def
aucts          0 0
UUID=3864a2c0-b796-454d-a6ac-2d359d76fb44 /boot             xfs      def
aucts          0 0
UUID=ebd9cb52-b111-44bb-a7ae-ed01b37bda27 none              swap     def
aucts          0 0
UUID=98ec12c6-6f15-4755-8e2c-24592f14d908 /srv/appdata xfs defaults 0 0
UUID=0fbffdf6-d7c3-47db-b0d0-fbfd82087898 /data2/app2 xfs defaults 0 0
[root@localhost ~]# umount /data2/app2
[root@localhost ~]# mount -a
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[root@localhost ~]# df -h /data2/app2
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/vgapp-lvapp  252M      21M   232M   9% /data2/app2
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                  8:0      0    15G  0 disk
├─sda1                8:1      0     1M  0 part
├─sda2                8:2      0    976M  0 part /boot
├─sda3                8:3      0    10G  0 part /
├─sda4                8:4      0    976M  0 part [SWAP]
├─sda5                8:5      0    400M  0 part
├─└─appvg-applv      253:0    0    352M  0 lvm  /srv/appdata
└─sda6                8:6      0    350M  0 part
   └─vgapp-lvapp      253:1    0    316M  0 lvm  /data2/app2
sr0                  11:0     1     51M  0 rom
[root@localhost ~]# █

```

Question 4: Cron Jobs and Log Monitoring A system administrator needs to monitor disk utilization every 30 minutes. Create a cron job that executes a script to record filesystem usage using `df -h` and appends the output with a timestamp to `/var/log/disk_monitor.log`. Ensure that the script and log file have appropriate permissions.

```

[root@localhost ~]# sudo vi /usr/local/bin/disk_monitor.sh
[root@localhost ~]# sudo chmod 755 /usr/local/bin/disk_monitor.sh
[root@localhost ~]# ls -l /usr/local/bin/disk_monitor.sh
-rwxr-xr-x. 1 root root 163 Oct  2 18:49 /usr/local/bin/disk_monitor.sh
[root@localhost ~]# sudo touch /var/log/disk_monitor.log
[root@localhost ~]# sudo chmod 644 /var/log/disk_monitor.log
[root@localhost ~]# ls -l /var/log/disk_monitor.log
-rw-r--r--. 1 root root 0 Oct  2 19:02 /var/log/disk_monitor.log
[root@localhost ~]# sudo /usr/local/bin/disk_monitor.sh
/usr/local/bin/disk_monitor.sh: line 1: in/bash: No such file or directory
[root@localhost ~]# sudo cat /usr/local/bin/disk_monitor.sh
in/bash

echo "===== $(date '+%Y-%m-%d %H:%M:%S') =====> /var/log/disk_monitor.log
df -h >> /var/log/disk_monitor.log
echo "" >> /var/log/disk_monitor.log
[root@localhost ~]# sudo crontab -e
crontab: installing new crontab
/root/.cache/crontab: mkdir: No such file or directory
[root@localhost ~]# sudo crontab -l

*/30 * * * * /usr/local/bin/security_check.sh
[root@localhost ~]# systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset:
   Active: active (running) since Fri 2026-10-02 11:17:07 IST; 8h ago
  Invocation: 525857ca7afd4eccb06d334e10f99188
    Main PID: 808 (crond)
      Tasks: 1 (limit: 22868)
     Memory: 1.3M (peak: 4M)
        CPU: 634ms
    CGroup: /system.slice/crond.service
            └─808 /usr/sbin/crond -n

Oct 02 17:01:01 localhost.localdomain CROND[5612]: (root) CMD (run-parts /et>
Oct 02 17:01:01 localhost.localdomain run-parts[5621]: (/etc/cron.hourly) fi>
Oct 02 17:01:01 localhost.localdomain CROND[5611]: (root) CMDEND (run-parts >
Oct 02 18:01:01 localhost.localdomain CROND[6446]: (root) CMD (run-parts /et>
Oct 02 18:01:01 localhost.localdomain run-parts[6449]: (/etc/cron.hourly) st>
Oct 02 18:01:01 localhost.localdomain CROND[6445]: (root) CMDEND (run-parts >
Oct 02 19:01:01 localhost.localdomain CROND[6964]: (root) CMD (run-parts /et>
Oct 02 19:01:01 localhost.localdomain run-parts[6967]: (/etc/cron.hourly) st>
Oct 02 19:01:01 localhost.localdomain CROND[6963]: (root) CMDEND (run-parts >
Oct 02 19:16:01 localhost.localdomain crond[808]: (root) RELOAD (/var/spool/>

```

Question 5: Networking and Service Availability A web application is hosted on a RHEL server with IP address 192.168.10.20/24. Configure a static IP address, gateway 192.168.10.1 and DNS server 8.8.8.8 using NetworkManager. Verify connectivity to the gateway and test DNS resolution.

```
[root@localhost ~]# nmcli connection show
```

NAME	UUID	TYPE	DE
enp0s3	cf290bda-687f-388d-80fe-e1b50a93b08d	ethernet	en
Wired connection 1	67262375-62dc-37a9-b501-da243248a425	ethernet	en
lo	934100ce-5888-4687-bea9-5b8acbf8ba23	loopback	lo

```
...skipping...

NAME                UUID                TYPE                DE
enp0s3              cf290bda-687f-388d-80fe-e1b50a93b08d  ethernet            en
Wired connection 1  67262375-62dc-37a9-b501-da243248a425  ethernet            en
lo                  934100ce-5888-4687-bea9-5b8acbf8ba23  loopback            lo
```

```
[root@localhost ~]# sudo nmcli connection modify enp0s3 ipv4.method manual
ipv4.addresses 192.168.10.20/24 ipv4.gateway 192.168.10.1 ipv4.dns 8.8.8.8
[root@localhost ~]# sudo nmcli connection up enp0s3
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
[root@localhost ~]# ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:43:c8:d2 brd ff:ff:ff:ff:ff:ff
    altname enx08002743c8d2
    inet 192.168.10.20/24 brd 192.168.10.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:a00:27ff:fe43:c8d2/64 scope global dynamic noprefixroute
        valid_lft 86367sec preferred_lft 14367sec
    inet6 fe80::a00:27ff:fe43:c8d2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost ~]# ip route
default via 192.168.31.1 dev enp0s8 proto dhcp src 192.168.31.196 metric 101
default via 192.168.10.1 dev enp0s3 proto static metric 102
192.168.10.0/24 dev enp0s3 proto kernel scope link src 192.168.10.20 metric 102
192.168.31.0/24 dev enp0s8 proto kernel scope link src 192.168.31.196 metric 101
```

```

[root@localhost ~]# ip neigh show dev enp0s3
192.168.10.1 FAILED
fe80::2 lladdr 52:54:00:12:35:00 router STALE
[root@localhost ~]# ping -c 4 192.168.10.20
PING 192.168.10.20 (192.168.10.20) 56(84) bytes of data.
64 bytes from 192.168.10.20: icmp_seq=1 ttl=64 time=0.087 ms
64 bytes from 192.168.10.20: icmp_seq=2 ttl=64 time=0.132 ms
64 bytes from 192.168.10.20: icmp_seq=3 ttl=64 time=0.177 ms
64 bytes from 192.168.10.20: icmp_seq=4 ttl=64 time=0.113 ms

--- 192.168.10.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3081ms
rtt min/avg/max/mdev = 0.087/0.127/0.177/0.032 ms

```

Question 6: Permissions, Logs and Security An application running under the user appuser writes logs to /opt/application/logs. Configure permissions so that appuser can create and modify log files, members of the support group can read them, and other users have no access. Configure a suitable log rotation policy that preserves seven compressed backup files.

```

[root@localhost ~]# groupadd support
[root@localhost ~]# mkdir -p /opt/application/logs
[root@localhost ~]# usermod -aG support appuser
[root@localhost ~]# chown -R appuser:support /opt/application/logs
[root@localhost ~]# chmod 2770 /opt/application/logs

[root@localhost ~]# touch /opt/application/logs/application.log
[root@localhost ~]# chown appuser:support /opt/application/logs/applica
tion.log
[root@localhost ~]# chmod 640 /opt/application/logs/*.log
[root@localhost ~]# ls -l /opt/application/logs/
total 0

[root@localhost ~]# cat /etc/logrotate.d/application
/opt/application/logs/*.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
    create 0640 appuser support
}

```

Question 7: Storage, Cron and Backup A server has a mounted backup directory at /backup. Create a shell script that archives /etc into a timestamped compressed archive

under /backup. Schedule the script to run every day at 1:00 AM using cron. Ensure that the backup files are accessible only to the root user.

```
[root@localhost ~]# vi /usr/local/bin/etc_backup.sh
[root@localhost ~]# cat /usr/local/bin/etc_backup.sh
#!/bin/bash

TIMESTAMP=$(date +%Y%m%d_%H%M%S)
BACKUP_FILE="/backup/etc_backup_${TIMESTAMP}.tar.gz"

tar -czf "$BACKUP_FILE" /etc
chmod 600 "$BACKUP_FILE"
[root@localhost ~]# chmod 700 /usr/local/bin/etc_backup.sh
[root@localhost ~]# ls -l /usr/local/bin/etc_backup.sh
-rwx-----. 1 root root 154 Oct  3 12:55 /usr/local/bin/etc_backup.sh
[root@localhost ~]# /usr/local/bin/etc_backup.sh
tar: Removing leading '/' from member names
[root@localhost ~]# ls -lh /backup/
total 5.6M
-rw-----. 1 root root 5.6M Oct  3 12:57 etc_backup_20261003_125739.tar.gz
drwxr-xr-x. 2 root root  38 Aug 24 20:05 myapp
drwxr-xr-x. 2 root root  30 Aug 24 15:33 projects
-rw-r--r--. 1 root root 195 Aug 24 14:53 projects_.tar.gz

[root@localhost ~]# crontab -e
crontab: installing new crontab
/root/.cache/crontab: mkdir: No such file or directory
[root@localhost ~]# ls -lh /backup/
total 5.6M
-rw-----. 1 root root 5.6M Oct  3 12:57 etc_backup_20261003_125739.ta
r.gz
drwxr-xr-x. 2 root root  38 Aug 24 20:05 myapp
drwxr-xr-x. 2 root root  30 Aug 24 15:33 projects
-rw-r--r--. 1 root root 195 Aug 24 14:53 projects_.tar.gz
[root@localhost ~]# crontab -l

0 1 * * * /usr/local/bin/etc_backup.sh
[root@localhost ~]# df -h /backup
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda3       10G   2.6G  7.4G  26% /
```

Question 8: Networking and Firewall Configuration A Linux server hosts an HTTP application on TCP port 8080. Configure the firewall to allow incoming connections to this port while keeping other unrequested services blocked. Verify the firewall configuration and test access to the application from another system.

```

[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-10-03 08:09:01 IST; 5h 52min ago
 Invocation: 244cbb5a04349b69e82f8dd29bcace8
    Docs: man:firewalld(1)
   Main PID: 738 (firewalld)
     Tasks: 2 (limit: 22868)
    Memory: 46.3M (peak: 46.6M)
       CPU: 1.645s
   CGroup: /system.slice/firewalld.service
           └─738 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Oct 03 08:09:00 localhost systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon
Oct 03 08:09:01 localhost systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
[root@localhost ~]# firewall-cmd --list-all
public (default, active)
  target: default
 ingress-priority: 0
 egress-priority: 0
 icmp-block-inversion: no
 interfaces: enp0s3 enp0s8
  sources:
 services: cockpit dhcpv6-client ssh
  ports:
 protocols:
 forward: yes
 masquerade: no
 forward-ports:
 source-ports:
 icmp-blocks:
 rich rules:
[root@localhost ~]# firewall-cmd --permanent --add-port=8080/tcp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-ports
8080/tcp

```

Question 9: Log Management and Cron-Based Cleanup A server generates temporary application logs under /var/log/application/temp. Create a scheduled task that deletes log files older than 15 days every Sunday at midnight. Before deletion, record the names and timestamps of the files selected for removal in a separate audit log. Ensure the cleanup task runs with appropriate privileges.

```

[root@localhost ~]# mkdir -p /var/log/application/temp
touch /var/log/application/temp_cleanup_audit.log
[root@localhost ~]# vi /usr/local/bin/cleanup_app_logs.sh
[root@localhost ~]# cat /usr/local/bin/cleanup_app_logs.sh
#!/bin/bash

LOG_DIR="/var/log/application/temp"
AUDIT_LOG="/var/log/application/temp_cleanup_audit.log"

find "$LOG_DIR" -type f -mtime +15 -printf '%TY-%Tm-%Td %TH:%TM:%TS %p\n' >> "$AUDIT_LOG"

find "$LOG_DIR" -type f -mtime +15 -delete
[root@localhost ~]# chmod 750 /usr/local/bin/cleanup_app_logs.sh
[root@localhost ~]# ls -l /usr/local/bin/cleanup_app_logs.sh
-rwxr-x---. 1 root root 240 Oct  3 15:05 /usr/local/bin/cleanup_app_log
s.sh
[root@localhost ~]# sudo /usr/local/bin/cleanup_app_logs.sh
[root@localhost ~]# cat /var/log/application/temp_cleanup_audit.log

#!/bin/bash

LOG_DIR="/var/log/application/temp"
AUDIT_LOG="/var/log/application/temp_cleanup_audit.log"

mkdir -p "$LOG_DIR"
touch "$AUDIT_LOG"
chmod 600 "$AUDIT_LOG"

find "$LOG_DIR" -type f -mtime +15 -printf '%TY-%Tm-%Td %TH:%TM:%TS %p\n' >> "$AUDIT_LOG"

find "$LOG_DIR" -type f -mtime +15 -delete
[root@localhost ~]# sudo crontab -e
crontab: installing new crontab
/root/.cache/crontab: mkdir: No such file or directory
[root@localhost ~]# sudo crontab -l

0 0 * * 0 /usr/local/bin/cleanup_app_logs.sh
[root@localhost ~]# find "$LOG_DIR" -type f -mtime +15 -printf '%TY-%Tm
-%Td %TH:%TM:%TS %p\n' >> "$AUDIT_LOG"
-bash: : No such file or directory
[root@localhost ~]# find "$LOG_DIR" -type f -mtime +15 -delete
find: `': No such file or directory
[root@localhost ~]# █

```

Question 10: Integrated Linux Administration Scenario A company hosts an internal application on a RHEL server. The application data is stored under /srv/appdata, logs are

stored under `/var/log/companyapp`, and the server uses a static IP address of `192.168.20.50/24`. Configure the environment to meet the following requirements:

- Create a dedicated `appadmin` user and `appteam` group, with appropriate directory permissions.
- Configure a 400 MB XFS logical volume mounted persistently at `/srv/appdata`.
- Configure log rotation to retain 14 days of compressed application logs.
- Schedule a daily backup of `/srv/appdata` at 2:00 AM using `cron`.
- Configure the firewall to allow SSH and HTTP traffic.
- Verify storage, permissions, network connectivity, scheduled tasks and log rotation.

```
[root@localhost ~]# fdisk /dev/sda5
```

Welcome to fdisk (util-linux 2.40.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/sda5: No such file or directory

```
[root@localhost ~]# fdisk /dev/sda
```

Welcome to fdisk (util-linux 2.40.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): n

Partition number (5-128, default 5): 5

First sector (24973312-31457246, default 24973312):

Last sector, +/-sectors or +/-size{K,M,G,T,P} (24973312-31457246, default 31455231): +400M

Created a new partition 5 of type 'Linux filesystem' and of size 400 MiB.

Command (m for help): p

Disk /dev/sda: 15 GiB, 16106127360 bytes, 31457280 sectors

Disk model: VBOX HARDDISK

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: gpt

Disk identifier: 5B3679E8-AA9A-4422-BDBF-A87850F47433

Device	Start	End	Sectors	Size	Type
/dev/sda1	2048	4095	2048	1M	BIOS boot
/dev/sda2	4096	2002943	1998848	976M	Linux extended boot
/dev/sda3	2002944	22974463	20971520	10G	Linux root (x86-64)
/dev/sda4	22974464	24973311	1998848	976M	Linux swap
/dev/sda5	24973312	25792511	819200	400M	Linux filesystem

Command (m for help): w

The partition table has been altered.

```

[root@localhost ~]# partprobe
Warning: Unable to open /dev/sr0 read-write (Read-only file system). /dev/sr
0 has been opened read-only.
[root@localhost ~]# sync
[root@localhost ~]# pvcreate /dev/sda5
Physical volume "/dev/sda5" successfully created.
[root@localhost ~]# vgcreate appvg /dev/sdb
No device found for /dev/sdb.
[root@localhost ~]# vgcreate appvg /dev/sda5
Volume group "appvg" successfully created
[root@localhost ~]# lvcreate -L 350M -n applv appvg
Rounding up size to full physical extent 352.00 MiB
Logical volume "applv" created.
[root@localhost ~]# lvs
LV      VG      Attr          LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync
Convert
applv   appvg   -wi-a----- 352.00m

[root@localhost ~]# mkfs.xfs /dev/appvg/applv
meta-data=/dev/appvg/applv      isize=512    agcount=4, agsize=22528 blks
=                               sectsz=512    attr=2, projid32bit=1
=                               crc=1        finobt=1, sparse=1, rmapbt=1
=                               reflink=1    bigtime=1 inobtcount=1 nnext64=
1
=                               exchange=0
data    =                               bsize=4096   blocks=90112, imaxpct=25
=                               sunit=0      swidth=0 blks
naming  =version 2                       bsize=4096   ascii-ci=0, ftype=1, parent=0
log     =internal log                   bsize=4096   blocks=16384, version=2
=                               sectsz=512    sunit=0 blks, lazy-count=1
realtime =none                         extsz=4096   blocks=0, rtextents=0
[root@localhost ~]# mkdir -p /srv/appdata
[root@localhost ~]# mount /dev/appvg/applv /srv/appdata
[root@localhost ~]# df -h /srv/appdata
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/appvg-applv 288M   24M  265M   9% /srv/appdata
[root@localhost ~]# blkid /dev/appvg/applv
/dev/appvg/applv: UUID="98ec12c6-6f15-4755-8e2c-24592f14d908" BLOCK_SIZE="512"
" TYPE="xfs"

```

```

[root@localhost ~]# lsblk -f
NAME FSTYPE FSVER LABEL          UUID                                FSAV
AIL FSUSE% MOUNTPOINTS
sda
├─sda1
├─sda2
│   xfs                3864a2c0-b796-454d-a6ac-2d359d76fb44    594
│   .6M    35% /boot
├─sda3
│   xfs                f902a1e3-4a9c-42dd-b32f-af27dd971fa4    7
│   .2G    27% /
├─sda4
│   swap      1
│               [SWAP]
├─sda5
│   │ LVM2_m LVM2
│   │               DYYbxI-rys0-Ckrl-FlnF-emvG-sall-kbWXuq
│   └─appvg-applv
│       xfs                98ec12c6-6f15-4755-8e2c-24592f14d908
sr0  iso966 Jolie VBox_GAs_7.2.14 2026-07-17-16-42-02-69

[root@localhost ~]# ^C
[root@localhost ~]# cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Tue Aug 11 12:09:27 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=f902a1e3-4a9c-42dd-b32f-af27dd971fa4 /                xfs      def
aaults          0 0
UUID=3864a2c0-b796-454d-a6ac-2d359d76fb44 /boot            xfs      def
aaults          0 0
UUID=ebd9cb52-b111-44bb-a7ae-ed01b37bda27 none             swap     def
aaults          0 0

[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# systemctl daemon-reload
[root@localhost ~]# umount /srv/appdata 2>/dev/null
[root@localhost ~]# mount -a
[root@localhost ~]# df -hT /srv/appdata
Filesystem                Type      Size  Used Avail Use% Mounted on
/dev/mapper/appvg-applv xfs        288M   24M  265M   9% /srv/appdata
[root@localhost ~]# findmnt /srv/appdata
TARGET SOURCE                FSTYPE OPTIONS
/srv/appdata
/dev/mapper/appvg-applv
xfs      rw,relatime,seclabel,attr2,inode64,logbufs=8

```

```
[root@localhost ~]# chown -R root:appteam /srv/appdata
[root@localhost ~]# chmod 2775 /srv/appdata
[root@localhost ~]# ls -ld /srv/appdata
drwxrwsr-x. 2 root appteam 6 Oct  5 19:59 /srv/appdata
[root@localhost ~]# id appadmin
uid=6675(appadmin) gid=6679(appadmin) groups=6679(appadmin),6683(appadmins)
[root@localhost ~]# mkdir -p /var/log/companyapp
[root@localhost ~]# chown root:appteam /var/log/companyapp
[root@localhost ~]# chmod 2775 /var/log/companyapp
[root@localhost ~]# ls -ld /var/log/companyapp
drwxrwsr-x. 2 root appteam 6 Oct  5 20:36 /var/log/companyapp
```

```
[root@localhost ~]# vi /etc/logrotate.d/companyapp
[root@localhost ~]# su root appteam
bash: appteam: No such file or directory
[root@localhost ~]# ^C
[root@localhost ~]# logrotate -d /etc/logrotate.d/companyapp
warning: logrotate in debug mode does nothing except printing debug messages!
    Consider using verbose mode (-v) instead if this is not what you want.

reading config file /etc/logrotate.d/companyapp
Reading state from file: /var/lib/logrotate/logrotate.status
Allocating hash table for state file, size 64 entries
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Handling 1 logs

rotating pattern: /var/log/companyapp/*.log after 1 days empty log files are
not rotated, (14 rotations), old logs are removed
switching euid from 0 to 0 and egid from 0 to 6698 (pid 6280)
considering log /var/log/companyapp/*.log
    log /var/log/companyapp/*.log does not exist -- skipping
switching euid from 0 to 0 and egid from 6698 to 0 (pid 6280)
```

```
considering log /var/log/companyapp/*.log
log /var/log/companyapp/*.log does not exist -- skipping
switching euid from 0 to 0 and egid from 6698 to 0 (pid 6280)
[root@localhost ~]# logrotate -f /etc/logrotate.d/companyapp
[root@localhost ~]# ls -lh /var/log/companyapp/
total 0
[root@localhost ~]# cat /etc/logrotate.d/companyapp
/var/log/companyapp/*.log {
    daily
    rotate 14
    compress
    delaycompress
    missingok
    notifempty
    copytruncate
    su root appteam
}
[root@localhost ~]# mkdir -p /backup/appdata
[root@localhost ~]# vi /usr/local/bin/backup_appdata.sh
[root@localhost ~]# cat /usr/local/bin/backup_appdata.sh
#!/bin/bash

DATE=$(date +%Y%m%d_%H%M%S)
BACKUP_DIR="/backup/appdata"
SOURCE="/srv/appdata"

mkdir -p "$BACKUP_DIR"

tar -czf "$BACKUP_DIR/appdata_${DATE}.tar.gz" "$SOURCE"
[root@localhost ~]# chmod 750 /usr/local/bin/backup_appdata.sh
[root@localhost ~]# /usr/local/bin/backup_appdata.sh
tar: Removing leading `/' from member names
[root@localhost ~]# ls -lh /backup/appdata/
total 4.0K
-rw-r--r--. 1 root root 128 Oct  5 21:17 appdata 20261005 211725.tar.gz
```

```

[root@localhost ~]# crontab -e
crontab: installing new crontab
/root/.cache/crontab: mkdir: No such file or directory
[root@localhost ~]# crontab -l

0 2 * * * /usr/local/bin/backup_appdata.sh
[root@localhost ~]# nmcli connection show

```

NAME	UUID	TYPE	DEVICE
enp0s3	cf290bda-687f-388d-80fe-e1b50a93b08d	ethernet	enp0s3
Wired connection 1	67262375-62dc-37a9-b501-da243248a425	ethernet	enp0s8
lo	b2959cb5-dab3-4dae-a5fc-f59a26d28b58	loopback	lo

```

[root@localhost ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:43:c8:d2 brd ff:ff:ff:ff:ff:ff
    altname enx08002743c8d2
    inet 192.168.10.20/24 brd 192.168.10.255 scope global noprefixroute enp0s
3
        valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:a00:27ff:fe43:c8d2/64 scope global dynamic noprefi
xroute
        valid_lft 86371sec preferred_lft 14371sec
    inet6 fe80::a00:27ff:fe43:c8d2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:e3:82:42 brd ff:ff:ff:ff:ff:ff
    altname enx080027e38242
    inet 192.168.31.196/24 brd 192.168.31.255 scope global dynamic noprefixro
ute enp0s8
        valid_lft 20980sec preferred_lft 20980sec
    inet6 2409:4090:2040:a804:e982:d410:ce0f:e99/64 scope global dynamic nopr
efixroute
        valid_lft 81104sec preferred_lft 81104sec
    inet6 fe80::c296:1d0e:edde:b5a7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost ~]# nmcli con mod "System enp0s3" ipv4.addresses 192.168.20.5

```

```

[root@localhost ~]# nmcli connection show
NAME                                UUID                                TYPE      DEVICE
enp0s3                             cf290bda-687f-388d-80fe-e1b50a93b08d ethernet  enp0s3
Wired connection 1                 67262375-62dc-37a9-b501-da243248a425 ethernet  enp0s8
lo                                 b2959cb5-dab3-4dae-a5fc-f59a26d28b58 loopback   lo
[root@localhost ~]# nmcli con mod "enp0s3" ipv4.addresses 192.168.20.50/24
[root@localhost ~]# nmcli -f NAME,DEVICE con show
NAME                                DEVICE
enp0s3                             enp0s3
Wired connection 1                 enp0s8
lo                                 lo
[root@localhost ~]# nmcli con mod "enp0s3" ipv4.gateway 192.168.20.1
[root@localhost ~]# nmcli con mod "enp0s3" ipv4.dns "8.8.8.8"
[root@localhost ~]# nmcli con mod "enp0s3" ipv4.method manual
[root@localhost ~]# nmcli con up "enp0s3"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
[root@localhost ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:43:c8:d2 brd ff:ff:ff:ff:ff:ff
    altname enx08002743c8d2
    inet 192.168.20.50/24 brd 192.168.20.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:a00:27ff:fe43:c8d2/64 scope global dynamic noprefixroute
        valid_lft 86390sec preferred_lft 14390sec
    inet6 fe80::a00:27ff:fe43:c8d2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e3:82:42 brd ff:ff:ff:ff:ff:ff
    altname enx080027e38242
    inet 192.168.31.196/24 brd 192.168.31.255 scope global dynamic noprefixroute enp0s8
        valid_lft 20260sec preferred_lft 20260sec

```

```

    inet6 2409:4090:2040:a804:e982:d410:ce0f:e99/64 scope global dynamic nopr
efixroute
    valid_lft 80441sec preferred_lft 80441sec
    inet6 fe80::c296:1d0e:edde:b5a7/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; pre>
   Active: active (running) since Mon 2026-10-05 13:42:54 IST; 9h ago
  Invocation: b2284c8f9b7d4cb68aa3e7a425a086e7
     Docs: man:firewalld(1)
   Main PID: 736 (firewalld)
     Tasks: 2 (limit: 22868)
    Memory: 44.8M (peak: 45.8M)
       CPU: 1.119s
    CGroup: /system.slice/firewalld.service
            └─736 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Oct 05 13:42:53 localhost systemd[1]: Starting firewalld.service - firewalld>
Oct 05 13:42:54 localhost systemd[1]: Started firewalld.service - firewalld>
lines 1-14/14 (END)...skipping...
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 13:42:54 IST; 9h ago
  Invocation: b2284c8f9b7d4cb68aa3e7a425a086e7
     Docs: man:firewalld(1)
   Main PID: 736 (firewalld)
     Tasks: 2 (limit: 22868)
    Memory: 44.8M (peak: 45.8M)
       CPU: 1.119s
    CGroup: /system.slice/firewalld.service
            └─736 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Oct 05 13:42:53 localhost systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon
Oct 05 13:42:54 localhost systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
~

```

```

[root@localhost ~]# systemctl enable --now firewalld
[root@localhost ~]# firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@localhost ~]# firewall-cmd --permanent --add-service=http
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   15G  0 disk
├─sda1                8:1    0    1M  0 part
├─sda2                8:2    0   976M  0 part /boot
├─sda3                8:3    0    10G  0 part /
├─sda4                8:4    0   976M  0 part [SWAP]
├─sda5                8:5    0   400M  0 part
└─┬appvg-applv       253:0    0   352M  0 lvm  /srv/appdata
sr0                  11:0    1    51M  0 rom

[root@localhost ~]# lvs
  LV      VG      Attr              LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync
Convert
  applv  appvg  -wi-ao----- 352.00m

[root@localhost ~]# df -h /srv/appdata
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/appvg-applv 288M   24M  265M   9% /srv/appdata
[root@localhost ~]# df -Th /srv/appdata
Filesystem                Type  Size  Used Avail Use% Mounted on
/dev/mapper/appvg-applv xfs   288M   24M  265M   9% /srv/appdata
[root@localhost ~]# ls -ld /srv/appdata
drwxrwsr-x. 2 root appteam 6 Oct  5 19:59 /srv/appdata
[root@localhost ~]# id appadmin
uid=6675(appadmin) gid=6679(appadmin) groups=6679(appadmin),6683(appadmins)
[root@localhost ~]# grep /srv/appdata /etc/fstab
UUID=98ec12c6-6f15-4755-8e2c-24592f14d908 /srv/appdata xfs defaults 0 0
[root@localhost ~]# mount -a
[root@localhost ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo

```

```

    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
   group default qlen 1000
   link/ether 08:00:27:43:c8:d2 brd ff:ff:ff:ff:ff:ff
   altname enx08002743c8d2
   inet 192.168.20.50/24 brd 192.168.20.255 scope global noprefixroute enp0s
3
    valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:a00:27ff:fe43:c8d2/64 scope global dynamic noprefi
xroute
    valid_lft 85970sec preferred_lft 13970sec
    inet6 fe80::a00:27ff:fe43:c8d2/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
   group default qlen 1000
   link/ether 08:00:27:e3:82:42 brd ff:ff:ff:ff:ff:ff
   altname enx080027e38242
   inet 192.168.31.196/24 brd 192.168.31.255 scope global dynamic noprefixro
ute enp0s8
    valid_lft 19701sec preferred_lft 19701sec
    inet6 2409:4090:2040:a804:e982:d410:ce0f:e99/64 scope global dynamic nopr
efixroute
    valid_lft 79924sec preferred_lft 79924sec
    inet6 fe80::c296:1d0e:edde:b5a7/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
[root@localhost ~]# ping -c 4 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
From 192.168.20.50 icmp_seq=1 Destination Host Unreachable
From 192.168.20.50 icmp_seq=2 Destination Host Unreachable
From 192.168.20.50 icmp_seq=3 Destination Host Unreachable
From 192.168.20.50 icmp_seq=4 Destination Host Unreachable

--- 192.168.20.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3108ms
pipe 3
[root@localhost ~]# ping -c 4 google.com
PING google.com (2404:6800:4000:101f::8b) 56 data bytes
64 bytes from lcbomo-in-f139.1e100.net (2404:6800:4000:101f::8b): icmp_seq=1
ttl=111 time=42.8 ms
64 bytes from lcbomo-in-f139.1e100.net (2404:6800:4000:101f::8b): icmp_seq=2
ttl=111 time=43.0 ms

```

```
64 bytes from lcbomo-in-f139.1e100.net (2404:6800:4000:101f::8b): icmp_seq=3
ttl=111 time=43.0 ms
64 bytes from lcbomo-in-f139.1e100.net (2404:6800:4000:101f::8b): icmp_seq=4
ttl=111 time=43.3 ms
```

```
--- google.com ping statistics ---
```

```
4 packets transmitted, 4 received, 0% packet loss, time 316lms
rtt min/avg/max/mdev = 42.797/43.007/43.257/0.164 ms
```

```
[root@localhost ~]# crontab -l
```

```
0 2 * * * /usr/local/bin/backup_appdata.sh
```

```
[root@localhost ~]# systemctl status crond
```

```
● crond.service - Command Scheduler
```

```
Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: >
```

```
Active: active (running) since Mon 2026-10-05 13:42:54 IST; 10h ago
```

```
Invocation: 338a27b4760745058d7c55316cd1d09d
```

```
Main PID: 809 (crond)
```

```
Tasks: 1 (limit: 22868)
```

```
Memory: 1.3M (peak: 4M)
```

```
CPU: 674ms
```

```
CGroup: /system.slice/crond.service
```

```
└─809 /usr/sbin/crond -n
```

```
Oct 05 18:01:01 localhost.localdomain CROND[4389]: (root) CMDEND (run-parts >
Oct 05 19:01:01 localhost.localdomain CROND[4971]: (root) CMD (run-parts /et>
Oct 05 20:01:01 localhost.localdomain CROND[5676]: (root) CMD (run-parts /et>
Oct 05 21:01:01 localhost.localdomain CROND[6348]: (root) CMD (run-parts /et>
Oct 05 21:01:02 localhost.localdomain CROND[6347]: (root) CMDEND (run-parts >
Oct 05 22:01:01 localhost.localdomain CROND[6820]: (root) CMD (run-parts /et>
Oct 05 22:01:01 localhost.localdomain CROND[6819]: (root) CMDEND (run-parts >
Oct 05 23:01:01 localhost.localdomain CROND[7253]: (root) CMD (run-parts /et>
Oct 05 23:01:01 localhost.localdomain CROND[7252]: (root) CMDEND (run-parts >
Oct 05 23:25:01 localhost.localdomain crond[809]: (root) RELOAD (/var/spool/>
...skipping...
```

```
● crond.service - Command Scheduler
```

```
Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
```

```
Active: active (running) since Mon 2026-10-05 13:42:54 IST; 10h ago
```

```
Invocation: 338a27b4760745058d7c55316cd1d09d
```

```
Main PID: 809 (crond)
```

```
Tasks: 1 (limit: 22868)
```

```
Memory: 1.3M (peak: 4M)
```

```
CPU: 674ms
```

```
CGroup: /system.slice/crond.service
```

```
└─809 /usr/sbin/crond -n
```

```
Oct 05 18:01:01 localhost.localdomain CROND[4389]: (root) CMDEND (run-parts /etc/cron.hourly)
Oct 05 19:01:01 localhost.localdomain CROND[4971]: (root) CMD (run-parts /etc/cron.hourly)
Oct 05 20:01:01 localhost.localdomain CROND[5676]: (root) CMD (run-parts /etc/cron.hourly)
Oct 05 21:01:01 localhost.localdomain CROND[6348]: (root) CMD (run-parts /etc/cron.hourly)
Oct 05 21:01:02 localhost.localdomain CROND[6347]: (root) CMDEND (run-parts /etc/cron.hourly)
Oct 05 22:01:01 localhost.localdomain CROND[6820]: (root) CMD (run-parts /etc/cron.hourly)
Oct 05 22:01:01 localhost.localdomain CROND[6819]: (root) CMDEND (run-parts /etc/cron.hourly)
Oct 05 23:01:01 localhost.localdomain CROND[7253]: (root) CMD (run-parts /etc/cron.hourly)
Oct 05 23:01:01 localhost.localdomain CROND[7252]: (root) CMDEND (run-parts /etc/cron.hourly)
Oct 05 23:25:01 localhost.localdomain crond[809]: (root) RELOAD (/var/spool/cron/root)
```

```
[root@localhost ~]# cat /etc/logrotate.d/companyapp
/var/log/companyapp/*.log {
    daily
    rotate 14
    compress
    delaycompress
    missingok
    notifempty
    copytruncate
    su root appteam
}
[root@localhost ~]# logrotate -d /etc/logrotate.d/companyapp
warning: logrotate in debug mode does nothing except printing debug messages!
Consider using verbose mode (-v) instead if this is not what you want.

reading config file /etc/logrotate.d/companyapp
Reading state from file: /var/lib/logrotate/logrotate.status
Allocating hash table for state file, size 64 entries
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Creating new state
Handling 1 logs

rotating pattern: /var/log/companyapp/*.log after 1 days empty log files are
not rotated, (14 rotations), old logs are removed
switching euid from 0 to 0 and egid from 0 to 6698 (pid 7748)
considering log /var/log/companyapp/*.log
log /var/log/companyapp/*.log does not exist -- skipping
switching euid from 0 to 0 and egid from 6698 to 0 (pid 7748)
[root@localhost ~]# logrotate -f /etc/logrotate.d/companyapp
[root@localhost ~]# ls -lh /var/log/companyapp/
total 0
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
[root@localhost ~]#
```